C1.1 Enzymes and metabolism

NAME	DATE	TOTAL
		_____ / 44

About 45 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 **DEFINE** metabolism. [C1.1.2] [1]

2 **STATE** one example each of an anabolic reaction and a catabolic reaction. [C1.1.3] [2]

3 **STATE** what is meant by the active site. [C1.1.4] [1]

4 **STATE** two factors that can denature an enzyme. [C1.1.7] [2]

5 **DEFINE** activation energy. [C1.1.10] [1]

Section B: Understanding

15 MARKS

6 **EXPLAIN** why cells need many different enzymes. [C1.1.2] [2]

7 **DESCRIBE** induced-fit binding. [C1.1.5] [2]

8 **EXPLAIN** the role of molecular motion in enzyme catalysis. [C1.1.6] [3]

9 **EXPLAIN** the effect of increasing substrate concentration on reaction rate. [C1.1.8] [3]

10 **EXPLAIN** how changing the shape of an active site affects specificity. [C1.1.7] [2]

11 **SKETCH** an energy graph for a reaction with and without an enzyme, labelling both activation energies. [3]

[C1.1.10]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [C1.1.8, C1.1.9]

Students used catalase from liver to break down hydrogen peroxide at different temperatures. They measured the oxygen collected.

TEMPERATURE / °C	10	20	30	40	50	60
OXYGEN IN 60 S / CM³	6	12	24	33	18	3

a **CALCULATE** the rate of reaction at 40°C. [2]

b **IDENTIFY** the optimum temperature within this range. [1]

c **EXPLAIN** the results between 10°C and 30°C. [2]

d **EXPLAIN** the results between 40°C and 60°C. [3]

e **STATE** two variables that must be controlled. [2]

13 [C1.1.9]

In another trial, oxygen volume rose like this:

TIME / S	0	10	20	30	40
OXYGEN / CM^3	0	8	14	17	18

a CALCULATE the initial rate, using the first 10 seconds. [2]

b EXPLAIN why the rate falls over time. [2]

Section D: Longer answer

8 MARKS

14 EXPLAIN how an enzyme catalyses a reaction, and how temperature and pH affect this. [C1.1.4–10] [8]